# Force & Motion: Newton's 3rd Law

# What is Force & Motion?

Force and motion are everywhere around us! Let's understand these two important concepts:

• Force is a push or pull that makes things move, stop, or change direction. When you push a door open or pull a drawer, you're using force!

• Motion is when something moves from one place to another. A rolling ball, a flying kite, or you walking to school – all are examples of motion.

Remember: Without force, there would be no motion!

# Meet Newton's 3rd Law

"For every action, there is an equal and opposite reaction."

This is Newton's Third Law of Motion! It means whenever you push or pull something, it pushes or pulls you back with the same amount of force, but in the opposite direction.

Imagine two friends on roller skates facing each other. When one friend pushes the other, both friends roll backward! The push goes both ways at the same time.

# Understanding Action-Reaction Pairs

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Action Force: When you push something, you apply a force in one direction. This is called the action force. For example, when you push a wall, your hands apply force toward the wall.

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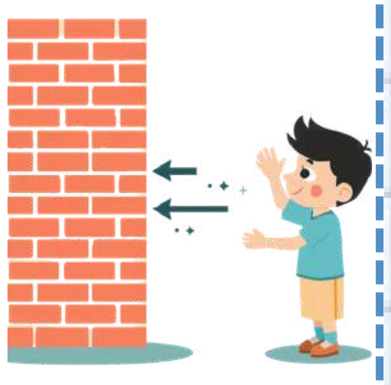
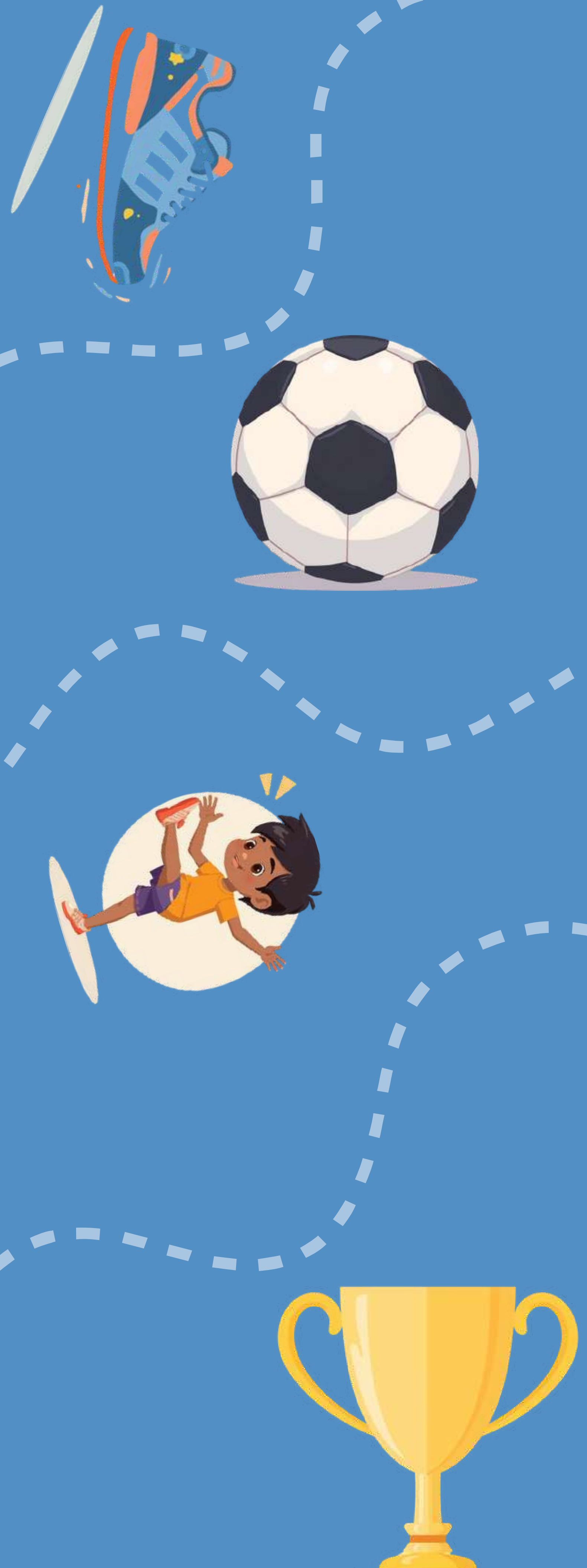
Reaction Force: The wall pushes back on your hands with equal force in the opposite direction! You can feel this push. Both forces happen at the same time on different objects.



## Real-World Example: Playground Push

Watch these children playing tug-of-war! When one team pulls the rope, the other team feels that pull and pulls back with equal force. This is Newton's 3rd Law in action!

The action force is one team pulling the rope toward themselves. The reaction force is the rope pulling back on their hands with equal strength. Both teams experience these paired forces at the same time!



## Real-World Example: Sports in Action

When you kick a soccer ball, Newton's 3rd Law is in action!
Action Force: Your foot pushes the ball forward with force.
Reaction Force: The ball pushes back on your foot with equal force in the opposite direction.
That's why your foot feels the impact when you kick! Both forces happen at the exact same moment.

## Real-World Example: Everyday Life

When you swim, your arms and legs push the water backward. This is the action force. The water then pushes you forward with an equal and opposite reaction force!

This is why swimmers move through the pool — by pushing water one way, the water pushes them the other way. Newton's 3rd Law is at work every time you take a stroke!

# Think About It!

**What happens when you let go of an inflated balloon?**

# Balloon Rocket Explained

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Action Force: When you release an inflated balloon, the air inside rushes out backward through the opening. This escaping air is the action force pushing in one direction.

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Reaction Force: As the air pushes backward, the balloon moves forward! This is the reaction force - equal in strength but opposite in direction. That's Newton's 3rd Law in action!

# Visual Demo: Walking

Every step you take demonstrates Newton's 3rd Law in action!

- Action Force: Your foot pushes backward against the ground
- Reaction Force: The ground pushes your foot forward

This is why you move forward when you walk! The harder you push against the ground, the stronger the ground pushes back, helping you walk faster or even run.

# Common Misconception #1

Misconception: "Action happens first, then reaction."
Many people think that one force happens before the other, like cause and effect.

The Truth: Action and reaction happen at the exact same time!
They always occur together as a pair — you cannot have one without the other.
When you push something, it pushes back instantly.

# Common Misconception #2

Misconception: "Equal and opposite forces cancel each other out, so nothing should move!"

The Truth: Action and reaction forces act on DIFFERENT objects, not the same one. When you push a cart, you push the cart (action) and the cart pushes you back (reaction). These forces don't cancel because they're on different things!

# Practice Problem #1

A skateboarder pushes against a wall. What forces are acting? Describe the action and reaction forces.

Think about it:

- What is the skateboarder doing to the wall?
- What does the wall do back to the skateboarder?
- Which direction does each force act?

Write your answer and discuss with your classmate!

# Practice Problem #2

Think about this scenario:

You are standing on a small boat near a dock. When you jump from the boat onto the dock, what happens to the boat?

Use Newton's 3rd Law to explain your answer. Think about:

- What is the action force?
- What is the reaction force?
- Which objects do these forces act on?

# Summary: Key Takeaways

Forces always come in pairs! Every time you push or pull something, it pushes or pulls back on you. This is true for every single interaction in the universe.

Action and reaction forces are always equal in strength and opposite in direction. When you push a wall with 10 Newtons, it pushes back with exactly 10 Newtons!

Action and reaction forces act on different objects, so they don't cancel out. This is why things can still move even though forces come in equal pairs.

# Thank You!

# Any Questions?